

Week 2_lab_Wireshark_Demo_and_assignment

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A1)

37	0.114494	10.79.2.168	10.3.0.43	TCP	66	51553 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
38	0.114664	10.79.2.168	10.3.0.43	TCP	66	59502 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
39	0.115109	10.79.2.168	10.3.0.43	TCP	66	55445 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
40	0.115255	10.79.2.168	10.3.0.43	TCP	66	54244 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
41	0.118713	10.3.0.43	10.79.2.168	TCP	66	53 → 51553 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1250 SACK_PERM WS=128
42	0.118713	10.3.0.43	10.79.2.168	TCP	66	53 → 55445 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1250 SACK_PERM WS=128
43	0.118713	10.3.0.43	10.79.2.168	TCP	66	53 → 54244 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1250 SACK_PERM WS=128
44	0.118940	10.79.2.168	10.3.0.43	TCP	54	51553 → 53 [ACK] Seq=1 Ack=1 Win=65280 Len=0
45	0.118986	10.79.2.168	10.3.0.43	TCP	54	55445 → 53 [ACK] Seq=1 Ack=1 Win=65280 Len=0
46	0.119000	10.79.2.168	10.3.0.43	TCP	54	54244 → 53 [ACK] Seq=1 Ack=1 Win=65280 Len=0

0.115255	54244 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM	53	TCP: 54244 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
0.118713	53 → 51553 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1250 SACK_PERM WS=128	53	TCP: 53 → 51553 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1250 SACK_PERM WS=128
0.118713	53 → 55445 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1250 SACK_PERM WS=128	53	TCP: 53 → 55445 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1250 SACK_PERM WS=128
0.118713	53 → 54244 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1250 SACK_PERM WS=128	53	TCP: 53 → 54244 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1250 SACK_PERM WS=128
0.118940	51553 → 53 [ACK] Seq=1 Ack=1 Win=65280 Len=0	53	TCP: 51553 → 53 [ACK] Seq=1 Ack=1 Win=65280 Len=0
0.118986	55445 → 53 [ACK] Seq=1 Ack=1 Win=65280 Len=0	53	TCP: 55445 → 53 [ACK] Seq=1 Ack=1 Win=65280 Len=0
0.119000	54244 → 53 [ACK] Seq=1 Ack=1 Win=65280 Len=0	53	TCP: 54244 → 53 [ACK] Seq=1 Ack=1 Win=65280 Len=0

A three-way handshake, is characterised by a triplet consisting of a SYN packet, a SYN, ACK packet, and an ACK packet.

Such a figure can be obtained by applying a basic TCP filter.

In the screenshot, a valid three-way handshake is visible, with a few re-transmissions mixed in.

Wireshark visually describes this in the packet info list, and via statistics->flow graph.

A2)

No.	Time	Source	Destination	Protocol	Length	Info
5	0.219442	10.79.2.168	148.113.8.188	TCP	55	57044 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
6	0.231237	148.113.8.188	10.79.2.168	TCP	66	443 → 57044 [ACK] Seq=1 Ack=2 Win=145 Len=0 SLE=1 SRE=2
14	0.359818	10.79.2.168	104.18.32.47	TCP	55	55154 → 443 [ACK] Seq=1 Ack=1 Win=255 Len=1
15	0.367750	104.18.32.47	10.79.2.168	TCP	66	443 → 55154 [ACK] Seq=1 Ack=2 Win=8249 Len=0 SLE=1 SRE=2

1	0.000000	10.79.6.137	10.79.15.255	UDP	86	57621 → 57621 Len=44
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Represented figures are obtained by

1. For TCP, opening any website
2. Running a DNS query such as `nslookup google.com`

TCP's are slower, connection-oriented, and have sequence numbers and Ack values.

UDP's are faster, connectionless, and lack sequence numbers and Ack values.

A3)

37	0.114494	10.79.2.168	10.3.0.43	TCP	66	51553 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
38	0.114664	10.79.2.168	10.3.0.43	TCP	66	59502 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
39	0.115109	10.79.2.168	10.3.0.43	TCP	66	55445 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
40	0.115255	10.79.2.168	10.3.0.43	TCP	66	54244 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
41	0.118713	10.3.0.43	10.79.2.168	TCP	66	53 → 51553 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1250 SACK_PERM WS=128
42	0.118713	10.3.0.43	10.79.2.168	TCP	66	53 → 55445 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1250 SACK_PERM WS=128
43	0.118713	10.3.0.43	10.79.2.168	TCP	66	53 → 54244 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1250 SACK_PERM WS=128
44	0.118940	10.79.2.168	10.3.0.43	TCP	54	51553 → 53 [ACK] Seq=1 Ack=1 Win=65280 Len=0
45	0.118986	10.79.2.168	10.3.0.43	TCP	54	55445 → 53 [ACK] Seq=1 Ack=1 Win=65280 Len=0
46	0.119000	10.79.2.168	10.3.0.43	TCP	54	54244 → 53 [ACK] Seq=1 Ack=1 Win=65280 Len=0

First SYN was at t1=0.114494

First ACK was at t3=0.118940

Handshake Duration=0.004446 seconds = 4.4ms

